FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ (SOLUTIONS)

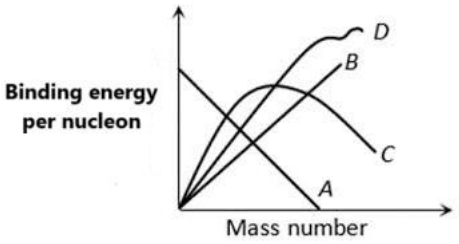
PHYSICS

CLASS XII

NUCLEI

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1. Binding energy per nucleon plot against the mass number for stable nuclei is shown in the figure. Which curve is correct?



(a) A (b) B (c) C (d) D

Ans: (c) C

2. A neutron can cause fission in

(a) hydrogen (b) uranium- 235 (c) thorium (d) uranium- 238

Ans: (b) uranium- 235

3. Order of magnitude of density of uranium nucleus [m_p= 1.67 x 10^{-27} kg] is

(a) 10^{20} kg/ m^3 (b) 10^{17} kg/ m^3 (c) 10^{14} kg/ m^3 (d) 10^{11} kg/ m^3

Ans: (b) 10^{17} kg/ m^3

4. Two spherical nuclei have mass numbers 216 and 64 with their radii R_1 and R_2 respectively. The ratio, $\frac{R1}{R2}$ is equal to

(a) 3:2 (b) 1:3 (c) 1:2 (d) 2:3

Ans: (a) 3:2

$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{216}{64}\right)^{1/3} = \frac{6}{4}$$

$$R_1 : R_2 = 3:2$$

5. In a nuclear reaction, there is conservation of

(a) mass only (b) energy only

(c) momentum only (d) mass, energy and momentum

Ans: (d) mass, energy and momentum

6. What is missing in the following nuclear reaction?

$${}_1^2H + {}_1^2H \longrightarrow {}_2^3He + ?$$

(a) meson (b) electron (c) positron (d) neutron

Ans: (d) neutron

7. What amount of energy is released by deuterium and tritium fusion?

(a) 60.6 MeV (b) 12.6 MeV (c) 17.6 MeV (d) 28.3 MeV

Ans: (c) 17.6 MeV

$$\underset{\text{Deuterium}}{{}_1^2\text{H}} + \underset{\text{Tritium}}{{}_1^3\text{H}} \longrightarrow \underset{\text{Helium}}{{}_2^4\text{He}} + \underset{\text{neutron}}{{}_0^1\text{n}} + Q$$

The energy released in the process is given by

$$Q = \left[M_{{}_1^2H} + M_{{}_1^3H} - M_{{}_2^4He} - M_{{}_0^1n}\right] c^2$$

$$= [2.014102 + 3.016050 - 4.002603 - 1.008665]\, uc^2$$

$$= (0.018884\, u)\left[931.5 \tfrac{MeV}{u}\right] = 17.6\, MeV.$$

8. The nuclei ${}_6^{13}C$ and ${}_7^{14}N$ can be described as
   (a) isotones (b) isobars
   (c) isotopes of carbon (d) isotopes of nitrogen
   Ans: (a) isotones
9. If the nucleus of ${}_{13}^{27}Al$ has a nuclear radius of about 3.6 fm, then ${}_{32}^{125}Te$ would have its radius approximately as
   (a) 9.6 fm (b) 12.0 fm (c) 4.8 fm (d) 6.0 fm
   Ans: (d) 6.0 fm

$$\frac{R_{\text{Te}}}{R_{\text{Al}}} = \left(\frac{A_{\text{Te}}}{A_{\text{Al}}}\right)^{1/3} = \left(\frac{125}{27}\right)^{1/3} = \left(\frac{5}{3}\right)$$

$$\text{or,}\quad R_{\text{Te}} = \frac{5}{3} \times R_{\text{Al}} = \frac{5}{3} \times 3.6 = 6 \text{ fm}$$

10. Which of the following statements is true for nuclear forces?
    (a) they obey the inverse square law of distances
    (b) they obey the inverse third power law of distance
    (c) they are short range forces
    (d) they are equal in strength to electromagnetic forces
    Ans: (c) they are short range forces
11. The mass of a proton is 1.0073 u and that of a neutron is 1.0087 u. The binding energy of ${}_2^4He$ is [Given helium nucleus mass = 4.0015 u]
    (a) 0.0305 J (b) 0.0305 erg (c) 28.4 MeV (d) 0.061 u
    Ans: (c) 28.4 MeV

$$\text{Mass defect} = 2M_P + 2M_N - M_{He}$$

$$= 2 \times 1.0073 + 2 \times 1.0087 - 4.0015 = 0.0305$$

$$\Rightarrow \text{Binding energy} = (931 \times \text{mass defect})\, MeV$$

$$= 931 \times 0.0305\, MeV = 28.4\, MeV$$

12. Fission of nuclei is possible because the binding energy per nucleon in them
    (a) increases with mass number at low mass numbers
    (b) decreases with mass number at low mass numbers

(c) increases with mass number at high mass numbers
(d) decreases with mass number at high mass numbers
Ans: (d) decreases with mass number at high mass numbers

13. Fusion reaction takes place at high temperatures because
(a) nuclei break up at high temperatures
(b) atoms get ionized at high temperatures
(c) kinetic energy is high enough to overcome the coulomb repulsion between nuclei
(d) molecules break up at high temperatures
Ans:(c) kinetic energy is high enough to overcome the coulomb repulsion between nuclei

For questions 14 to 18, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

14. **Assertion:** Nuclei having mass number about 60 are most stable.
Reason: When two or more light nuclei are combined into a heavier nucleus, then the binding energy per nucleon will increase.
Ans: B
15. **Assertion:** Density of all the nuclei is same.
Reason: Radius of nucleus is directly proportional to the cube root of mass number.
Ans: A
16. **Assertion:** Binding energy per nucleon is nearly constant for element in the range A=30 to A=170.
Reason: The nuclear force between two nucleons falls rapidly to zero as their distance is more than a few femtometres.
Ans: A
17. **Assertion:** It is not possible to use ^{35}Cl as the fuel for fusion energy.
Reason: The binding energy of ^{35}Cl is too small.
Ans: C

In fusion, lighter nuclei are used so, fusion is not possible with ^{35}Cl. Also binding energy of ^{35}Cl is not too small.

18. **Assertion:** Bohr model cannot be extended to two or more electron atoms.
Reason: Each electron in the atom interacts not only with the positively charged nucleus but also with all other electrons.
Ans: A